

Homework 08 - Poles, zeros of continuous systems & z -transform basics

Deadline: 3rd test date (14th of May)

The mandatory questions account for 100 points, while the optional questions together contribute up to an additional 50 points.

- Write a Python script using `sympy` to compute the Laplace transform of $f(t) = e^{-2t}$ and $g(t) = \sin(3t)$. Calculate their convolution and prove the validity of the convolution theorem for its Laplace transform.
- Define poles and zeros in the context of the Laplace transform. How do they influence the behavior of a system?
- Discuss the significance of the pole-zero plot in analyzing the stability of a system.
- Explain why all-pole analogue filters can only be low-pass or band-pass.
- What kind of filter does the transfer function $H(s) = \frac{s}{s+1}$ represent? Argue it using words instead of calculations.
- What is a Bode plot, and how is it used to analyze continuous-time systems? What are the key components of a Bode plot?
- Explain how to interpret the gain and phase plots in a Bode diagram.
- Compute and plot the Bode plot for the transfer function $H(s) = \frac{s+1}{s^2+3s+2}$. Comment each step to explain the code.
- Using Python, create a function to compute and plot the frequency response of an LTI system given its transfer function $H(s)$. Compare the results with those obtained from a Bode plot.
- What is the z -transform, and how does it differ from the Fourier transform?
- Explain the region of convergence (ROC) for the z -transform. Why is it important?
- How does the z -transform of the convolution of two sequences relate to the individual z -transforms of the two sequences?
- Prove the following properties of the z -transform of two sequences $x[n]$ and $y[n]$:

(a) Linearity:

$$\mathcal{Z}\{ax[n] + by[n]\} = aX(z) + bY(z)$$

(b) Time shifting:

$$\mathcal{Z}\{x[n-k]\} = z^{-k}X(z), \quad k \in \mathbb{Z}$$

(c) Scaling in the z -domain (modulation in time):

$$\mathcal{Z}\{a^n x[n]\} = X\left(\frac{z}{a}\right), \quad a \neq 0$$

(d) Time reversal:

$$\mathcal{Z}\{x[-n]\} = X(z^{-1})$$

(e)  Convolution:

$$\mathcal{Z}\{x[n] * y[n]\} = X(z)Y(z)$$

- Calculate the z -transform of the following sequences and determine their region of convergence:

(a) $u[n] = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$

(b) $\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$

- Compute the z -transform of the sequence $x[n] = a^n u[n]$. Verify your result by checking the region of convergence (ROC).

- Show that the z -transform of $x[n] = \delta[n] - \delta[n-1]$ is $X(z) = 1 - z^{-1}$.

- Determine the inverse z -transform of $X(z) = \frac{z}{z-a}$ for different regions of convergence and explain the steps involved.

- Show that if sampled at intervals T , the z -transform of the functions below are:

(a) $t \rightarrow \frac{Tz}{(z-1)^2}$

(b) $t^2 \rightarrow \frac{T^2 z(z+1)}{(z-1)^3}$

(c) $e^{-at} \rightarrow \frac{z}{z-e^{-aT}}$

(d)  $te^{-at} \rightarrow \frac{Tze^{-aT}}{(z-e^{-aT})^2}$

(e)  $\sin(\omega t) \rightarrow \frac{z \sin(\omega T)}{z^2 - 2z \cos(\omega T) + 1}$

(f)  $\cos(\omega t) \rightarrow \frac{z(z - \cos(\omega T))}{z^2 - 2z \cos(\omega T) + 1}$

- If the input to a system, having a transfer function of $1 + 2z^{-1} - z^{-2}$, is a discrete unit step, show that the first four terms of the output sequence are 1, 3, 2, 2.

- When a discrete signal 1, -2 is inputted to a processing system, the output sequence is 1, -5, 8, -4.

(a) Derive the transfer function for the system.

(b)  Find the first three terms of the output sequence in response to the finite input sequence of 2, 2, 1.

-  Implement a Python script to compute the z -transform of $x[n] = (0.5)^n u[n]$ and verify the region of convergence by visualizing a heat map of the z -transform for a dense grid of complex z values.